

Wasserstein distance is a metric for measuring distances between two probability measures.
(probability distributions)

1. non-negative $W(\mu, \nu) \geq 0$ where " $=$ " holds iff
2. Symmetric $W(\mu, \nu) = W(\nu, \mu)$
3. triangle inequality $W(\mu, \nu) + W(\nu, \eta) \geq W(\mu, \eta)$

↪ equivalent, since they uniquely decide each other

1. Definition

It is easier to start with discrete distributions. Suppose we have two discrete measurements μ and ν with finite support

$X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_m\}$, respectively.

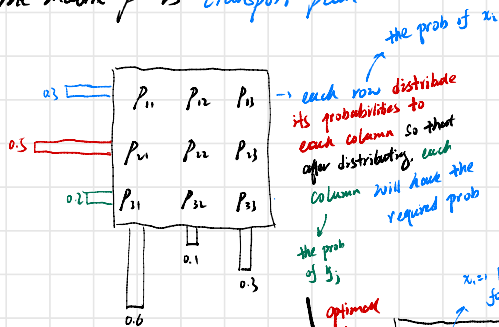
$C_{ij} = C(x_i, y_j)$ is a cost between x_i and y_j , e.g. L_2 loss.
then the Wasserstein distance between μ and ν can be defined as:

$$W(\mu, \nu) = \min_p \sum_{i=1}^n \sum_{j=1}^m p_{ij} C_{ij}$$

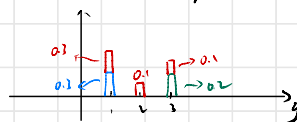
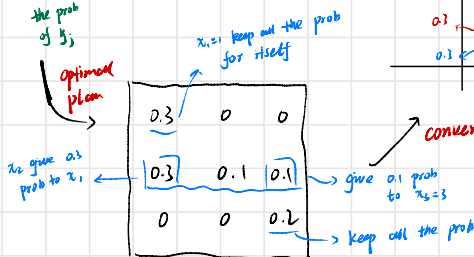
$$\text{s.t. } \sum_{j=1}^m p_{ij} = \mu(x_i) \text{ for any } i=1, \dots, n$$

$$\sum_{i=1}^n p_{ij} = \nu(y_j) \text{ for any } j=1, \dots, m$$

The matrix p is transport plan



$W(p, q)$: the lowest cost to move $p \rightarrow q$



If $C(x_i, y_j) = (x_i - y_j)^2$, then the Wasserstein distance here is

$$W(\mu, \nu) = 0.3 \cdot 0 + 0.3 \cdot 1 + 0.1 \cdot 0 + 0.1 \cdot 1 + 0.2 \cdot 0$$

$$= 0.4$$

2. Solution: Linear programming

We can formulate the problem for solving the Wasserstein distance between two discrete distributions as a **linear programming** problem:

$$\begin{aligned} \min_x \quad & C^T x \\ \text{s.t.} \quad & Ax = \beta \\ & x \geq 0 \end{aligned}$$

$$C = \begin{bmatrix} c_{1,1} \\ c_{1,2} \\ \vdots \\ c_{1,m} \\ \vdots \\ c_{n,1} \\ c_{n,2} \\ \vdots \\ c_{n,m} \end{bmatrix} \begin{array}{l} \rightarrow \text{first row} \\ \vdots \\ \rightarrow \text{last row} \end{array}$$

$$x = \begin{bmatrix} p_{1,1} \\ p_{1,2} \\ \vdots \\ p_{1,m} \\ \vdots \\ p_{n,1} \\ p_{n,2} \\ \vdots \\ p_{n,m} \end{bmatrix} \begin{array}{l} \rightarrow \text{first col} \\ \vdots \\ \rightarrow \text{last col} \end{array}$$

$$A_1 = \begin{pmatrix} \overbrace{1_{n \times n} \quad 0_{1 \times n} \quad \dots \quad 0_{1 \times n}}^{n \times m} \\ \vdots \\ 0_{1 \times n} \quad 1_{1 \times n} \quad \dots \quad 0_{1 \times n} \\ \vdots \\ 0_{1 \times n} \quad 0_{1 \times n} \quad \dots \quad 1_{1 \times n} \end{pmatrix} \left. \vphantom{\begin{pmatrix} 1_{n \times n} \\ \vdots \\ 0_{1 \times n} \end{pmatrix}} \right\} m$$

$$A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \quad \beta = \begin{pmatrix} v(y_1) \\ \vdots \\ v(y_m) \\ \mu(x_1) \\ \vdots \\ \mu(x_n) \end{pmatrix}$$

$$A_2 = \left(\underbrace{I_{n \times n} \quad I_{n \times n} \quad \dots \quad I_{n \times n}}_{n \times m} \right)$$

1. **Convex problem** A linear programming problem is guaranteed to be a convex problem, so local minimum $\Rightarrow$ global minimum

Why is it written as a linear programming problem useful?

2. **Algorithms for solving the problem.** There are many algorithms for it, e.g. Simplex method or interior point method. Although the complexity could be high.

3. **Strong duality holds**

3. Continuous probability measures

$$W(\mu, \nu) = \min_{\pi(\mu, \nu)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

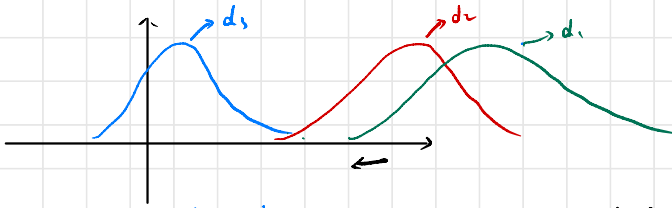
$$\text{s.t.} \quad \int_X \pi(x, y) dx = \nu(y)$$

$$\int_Y \pi(x, y) dy = \mu(x)$$

$\rightarrow$ i.e. $\pi(x, y)$ is a coupling of $\mu(x)$ and $\nu(y)$

4. Compare KL divergence with Wasserstein distance

KL divergence



$$KL(d_1, d_2) \approx KL(d_2, d_1)$$

It fails to capture the shift of distribution d_1 to d_2

$$W(d_1, d_2) > W(d_2, d_1)$$

It can capture the shift even if the supports are disjoint